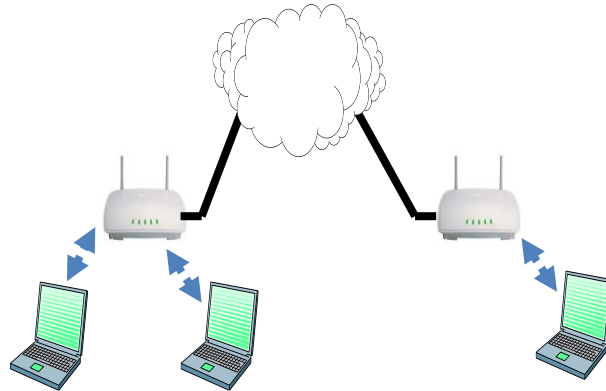


EEU44C04 / CS4031 / CS7NS3 / EEP55C27
Next Generation Networks

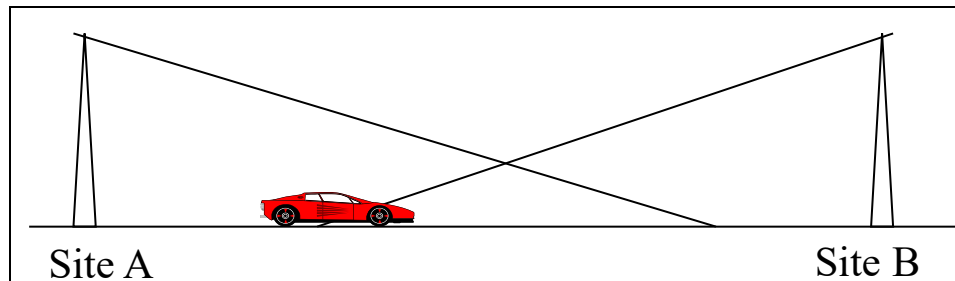
Overview of wireless networks

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Types of wireless networks



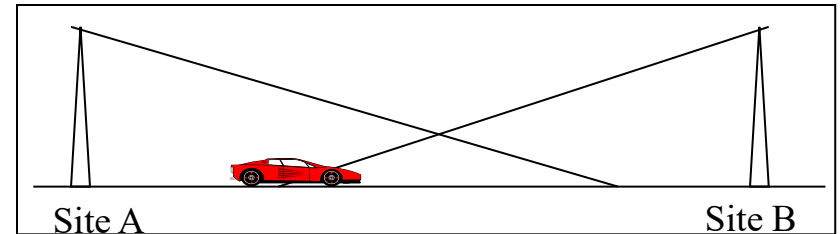
Nomadic



Mobile

Mobile

- Seamless roaming
- Wireless telephony
- Mobile data services
 - ✓ Email, web browsing, short message service (SMS), videos, social content
- Vehicular ad hoc networks
- Location-aware services



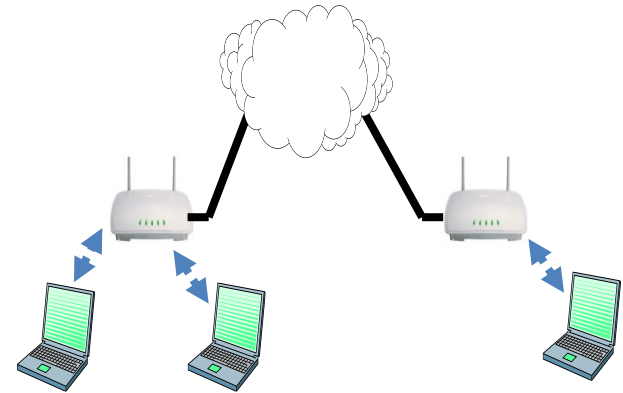
Technologies:

- Cellular/PCS (GSM, GPRS, UMTS, LTE, 5G, B5G/6G, etc.)
- IEEE 802.16
- Land mobile radio
- Satellite

Mobile (移动网络): 用户在移动中 (比如开车) 持续保持连接, 网络需要在你从一个基站移动到另一个基站时自动"接力" (即"切换/handover"), 做到无缝漫游。典型技术: 蜂窝网络 (GSM、4G、5G等)。

Nomadic

- User changes point of connection to the network
 - ✓ Seamless roaming not necessary
- Wireless local area networks (WLAN)
- Hot spots
- Wireless ISPs



Technologies:

- IEEE 802.11
- Bluetooth

Nomadic (游牧式/暂驻式网络)：用户会更换接入点,但不要求"无缝漫游"—比如你带笔记本电脑从家里的 WiFi 换到咖啡厅的 WiFi,中途断一下连接是可以接受的。典型技术:WLAN(802.11)、蓝牙、公共热点。

这是把经典的网络分层模型套到无线场景, 每一层负责的事情有些不同:

OSI reference model

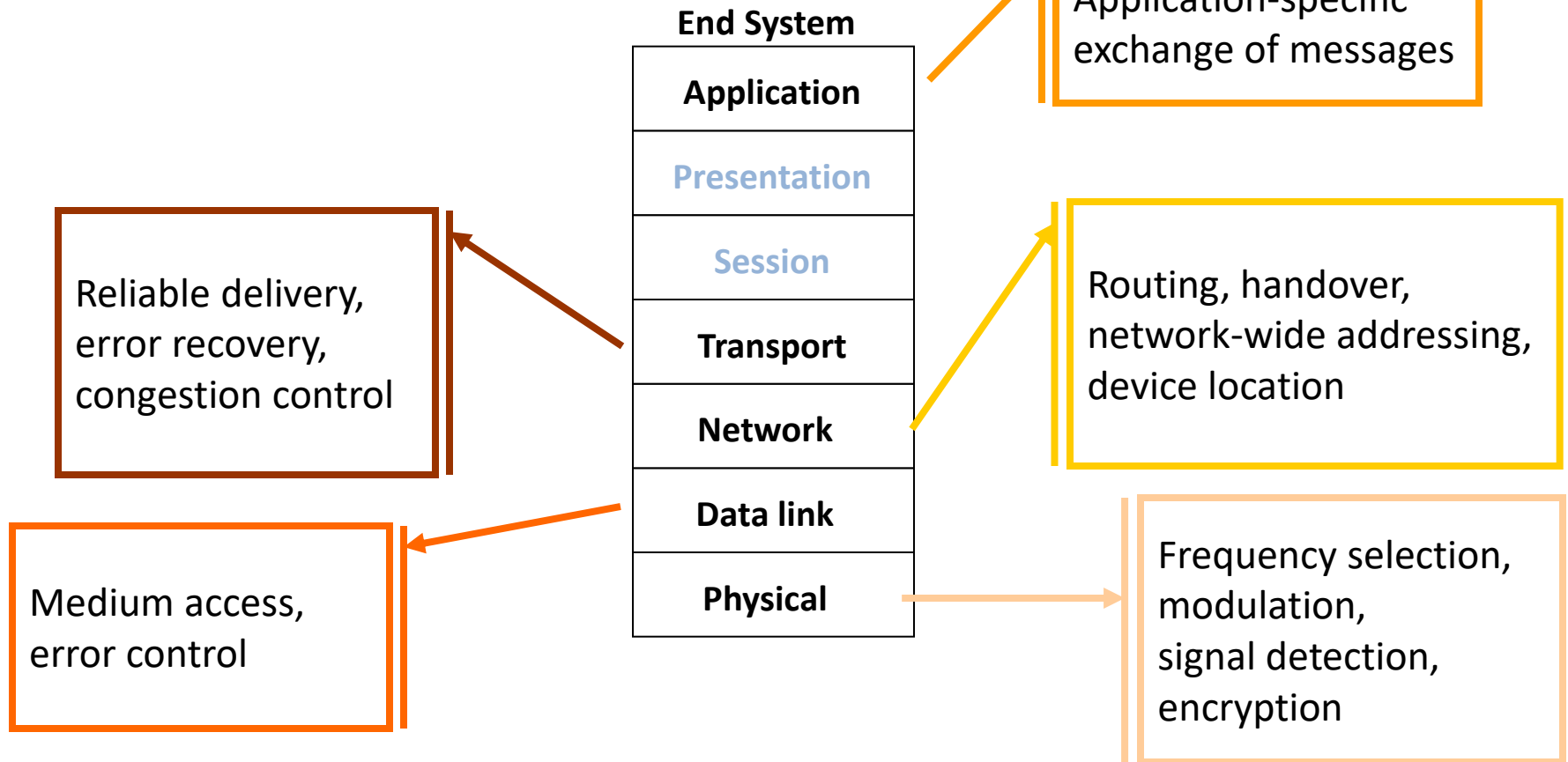
物理层(Physical): 处理频率选择、调制、信号检测、加密,以及应对噪声、干扰、衰落等无线信道特有的问题。

数据链路层(Data link): 管理谁能访问无线信道(媒体访问控制),做差错控制。

网络层(Network): 处理路由、基站间切换(handover)、设备定位。

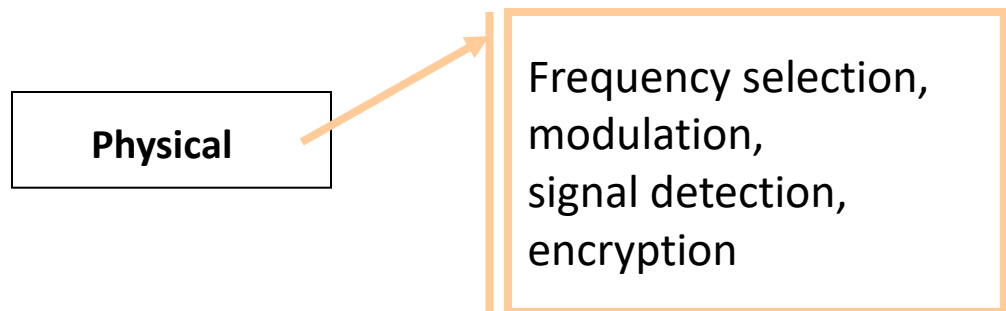
传输层(Transport): 保证端到端可靠传输、流量/拥塞控制。

应用层(Application): 针对移动设备的限制(小屏幕、低电量等)做优化,比如基于位置的服务。



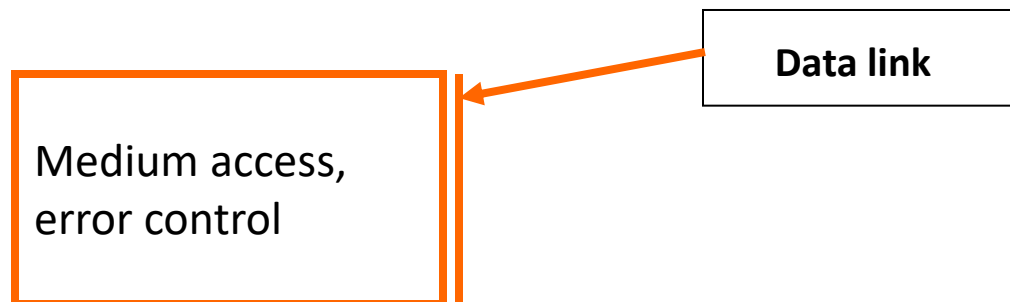
Physical layer (PHY)

- Handles **communications impairments** (noise, interference, fading, shadowing, path loss, etc.) through coding, diversity, power control, waveform selection, etc.
- Responsible for modulation, frequency selection, signal detection
- Encryption



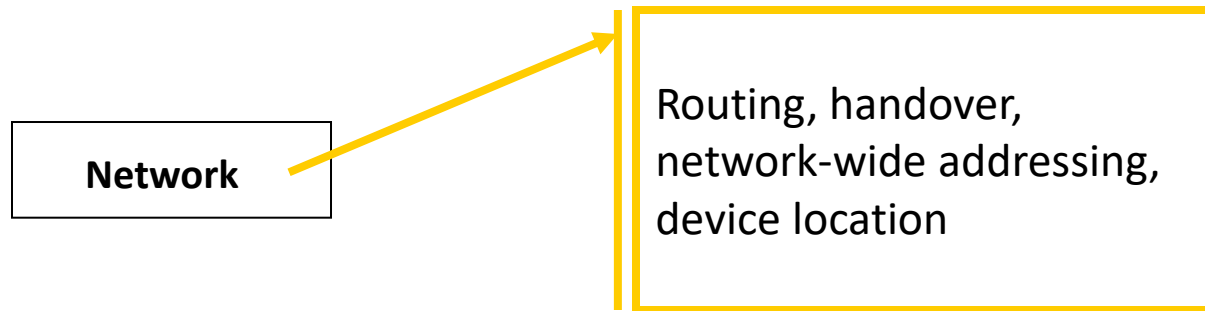
Data link layer

- Responsible for **mediating access** to the wireless medium
- Frame synchronization
- Reliable point-to-point or point-to-multipoint connection



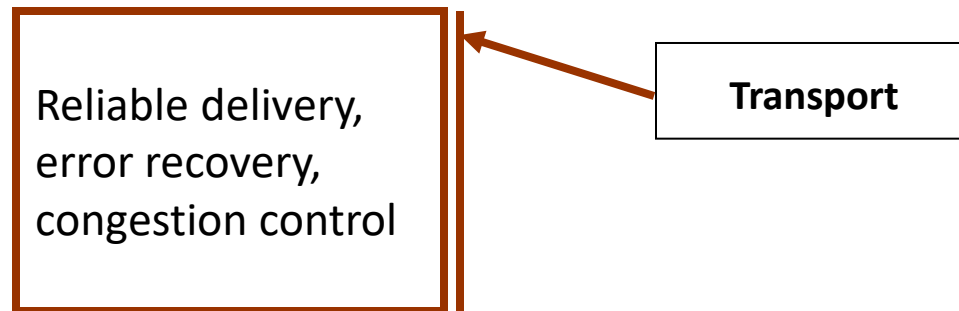
Network layer

- **Network-wide** addressing, and redirection to deal with mobility
- Handoff (handover) between different networks
- Routing through multiple hops in ad-hoc networks



Transport layer

- Reliable **end-to-end** transmission
- Flow and congestion control

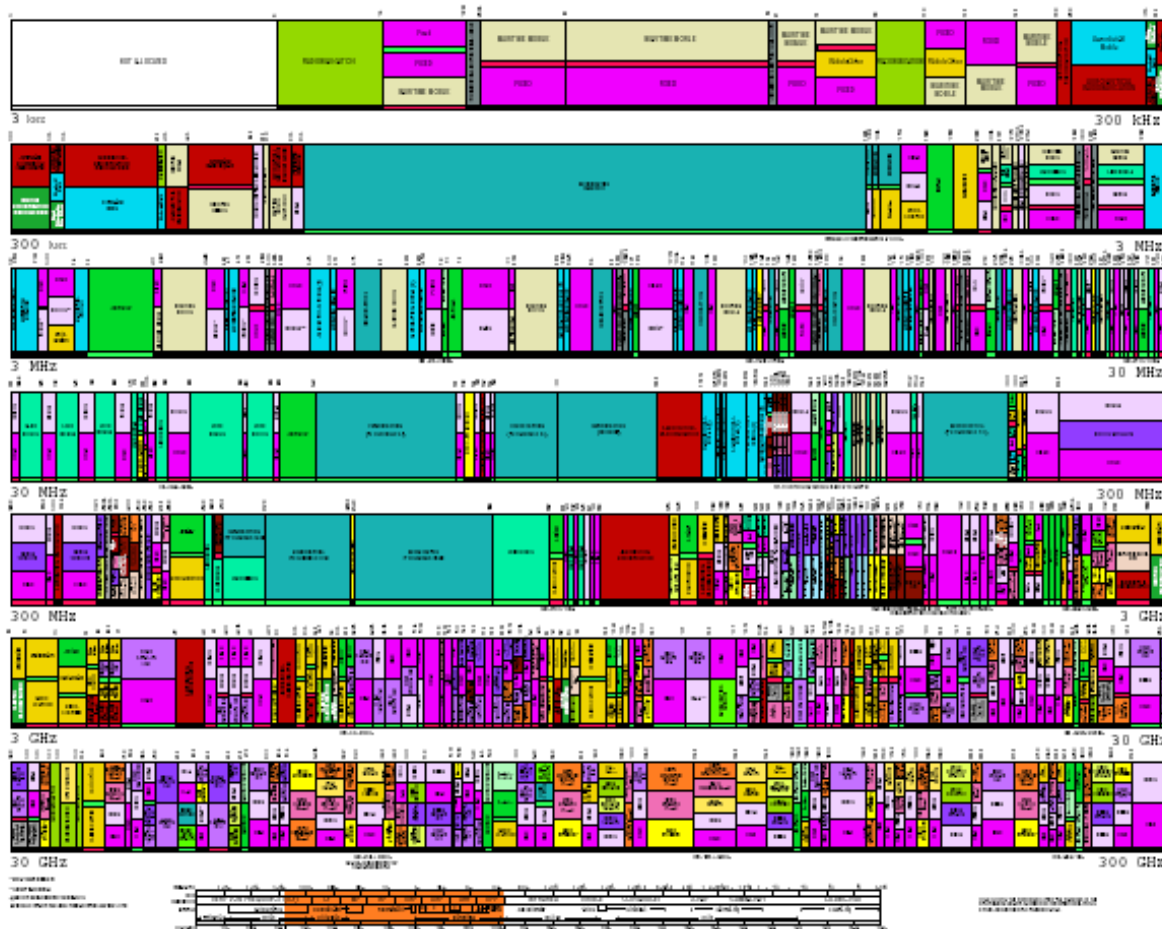


Application layer

- Application layer protocols may need to be tuned to support the constraints of **mobile devices** (small screen, low power, ...)
- Location-aware services
- Wireless web access



United States frequency allocations



Is it tidy?

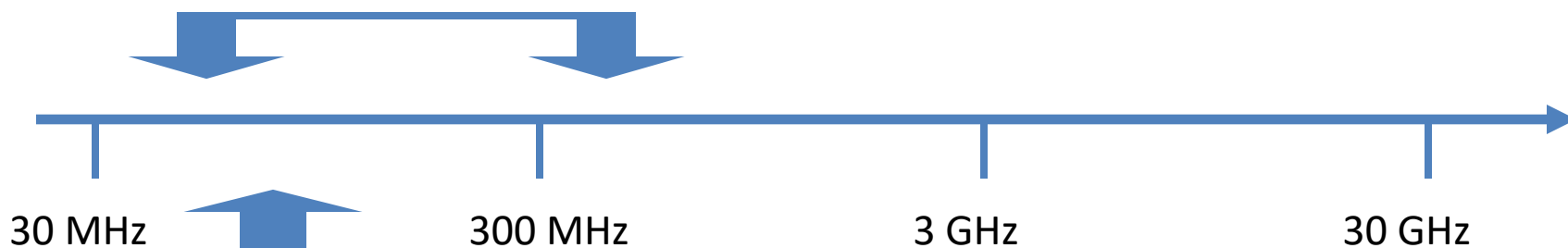
And by the way, what's up with the number 3? (3 KHz, 3 MHz etc.)

Broadcast radio and TV



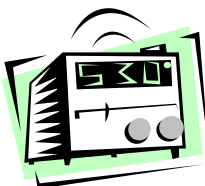
Broadcast TV

- VHF: 54 to 88 MHz, 174 to 216 MHz
- UHF: 470 to 806 MHz



FM Radio

- 88 to 108 MHz



Digital TV

- 54 to 88 MHz, 174 to 216 MHz, 470 to 806 MHz

这部分讲的是无线电频谱(3kHz 到 300GHz)是如何被切分给不同用途的:

广播电视和 FM 广播用较低频段(几十到几百 MHz)

蜂窝电话(2G/3G/4G/5G)用几百 MHz 到几 GHz

WiFi、蓝牙用 2.4GHz、5GHz

老师吐槽了一句"频谱分配图乱不乱?"以及"为什么总是用 3 这个数字做分界(3kHz, 3MHz, 3GHz)"

这只是每次乘以1000做单位换算的习惯(kHz→MHz→GHz), 没有特别深意,更多是在引导你观察规律。

Cellular and PCS



3G Broadband Wireless

- 746-794 MHz, 1.7-1.85 GHz, 2.5-2.7 GHz

4G

- 800 MHz, 1.8 GHz



2G Cellular Phone

- 800-900 MHz



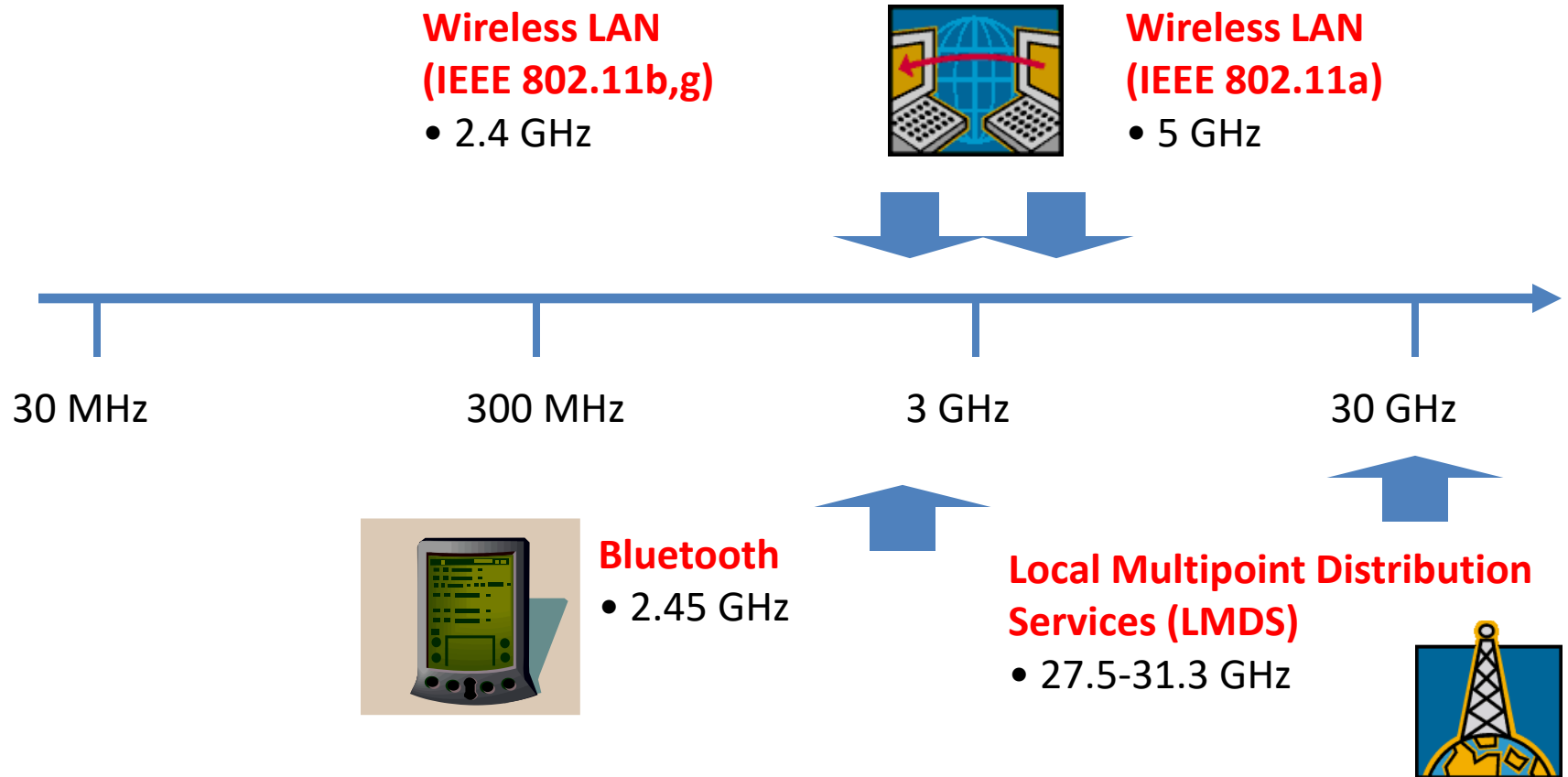
Personal Communication Service (PCS)

- 1.85-1.99 GHz

5G

- 3.4-3.8 GHz

WLANs, WPANs, fixed wireless



监管机构

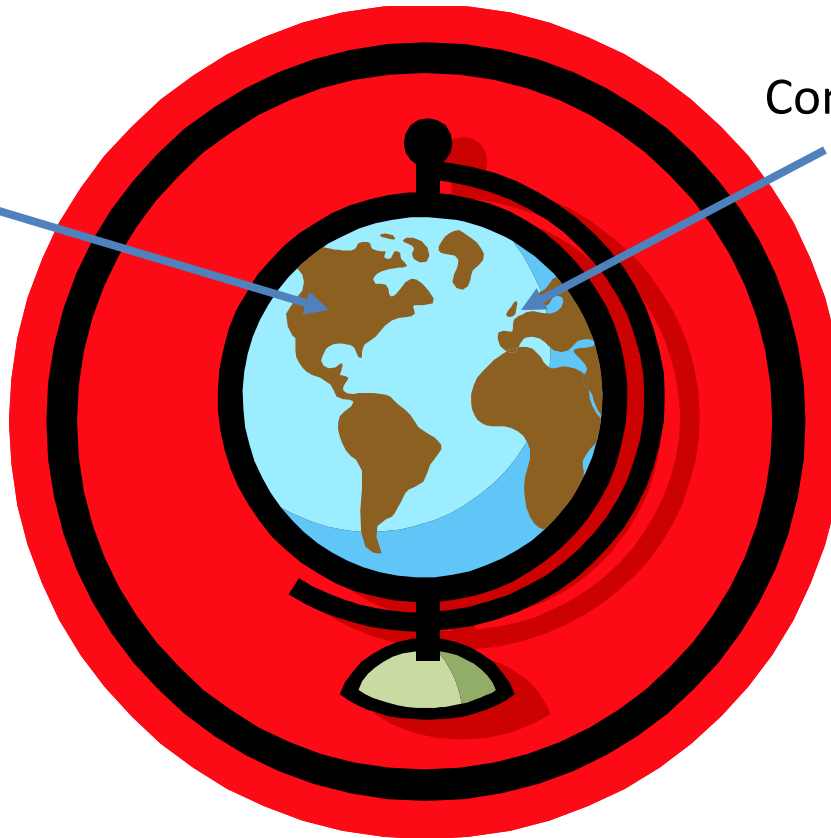
Regulators

ITU-R(国际电信联盟无线电通信部门):全球层面负责标准化和频率规划。
各国都有自己的监管机构,比如美国的 FCC,爱尔兰的 ComReg。

International Telecommunications Union – Radiocommunication sector
(ITU-R) handles **standardization and frequency planning**

Federal Communications
Commission (USA)

ComReg (Ireland)



ComReg = Commission for Communications Regulation

Harmonization(协调统一): 全球统一频谱分配虽然理想,但因为历史遗留系统太多很难做到,不过出于避免干扰和规模经济的考虑,还是有必要尽量协调(比如 ITU-R 定期召开世界无线电大会来决定频谱分配)。

Harmonization

- Some degree of harmonization (more rational/efficient re-organization) of spectrum utilization world-wide is desirable
 - ✓ Not easy: a lot of legacy systems (see former slide on US frequency allocations)
 - ✓ But necessary, both for interference avoidance (inter-operability) and for economic reasons (economies of scale)
 - ✓ ITU-R periodically holds the **World Radio Conference**, where the allocation of the spectrum to different services is decided

动态频谱接入(Dynamic Spectrum Access): 现有频谱利用率其实不高,但很难把已授权的频段收回。于是出现了"只要不干扰持牌用户,就可以借用其频段"的思路,认知无线电(Cognitive Radio)等自组织网络技术就是想实现这个目标。

Dynamic spectrum access

- Spectrum is not used very efficiently
 - ✓ But it is very difficult to take spectrum back from those licensed to operate in a given band
 - ✓ Counterexample: Spectrum re-farming of TV bands due to efficiencies in the switch to digital TV
 - ✓ **Dynamic spectrum access**: allows operation in licensed bands, as long as you don't interfere with holder of the license
 - Autonomous networks (e.g., Cognitive radio, Self-organizing networks, Zero Touch Networks) attempting to make this a reality

Which of the following regions of the spectrum is best suited and in fact used for TV transmission?

- ☐ $< 1\text{GHz}$
- ☐ Between 1GHz and 5GHz
- ☐ $>5\text{GHz}$



Which of the following regions of the spectrum is best suited and in fact used for TV transmission?

☒ $< 1\text{GHz}$

☐ Between 1GHz and 5GHz

☐ $> 5\text{GHz}$

题目一: 电视信号该用哪个频段?答案是 低于1GHz,因为频率越低,信号随距离衰减得越慢,适合远距离传输(这就是电视/广播用较低频段的原因)。

Transmissions at lower frequencies attenuate less with distance, which makes these bands ideal for long range transmissions, such as TV ones.

Comment on the complementarity of wired and wireless systems.



题目二(关于有线 vs 无线的互补性): 光纤覆盖范围有限,但覆盖到的地方带宽极大;无线网络几乎能覆盖任何地方,但带宽有限、容易受干扰——两者是互补关系,不是竞争关系。

- Wireless and optical wireless networks can be thought of as complementary.
- Optical fibre does not reach everywhere, but where it does reach, it provides a very large amount of bandwidth.
- Wireless access networks, on the other hand, potentially reach almost everywhere, but provide a highly bandwidth-constrained transmission channel, susceptible to a variety of impairments.

Given that the user in a certain mobile communication system, is moving with a velocity of 1,000 km/h, which of the following downlink power allocation strategies makes more sense, assuming multiple antennas at the base station?

- (i) The transmit power budget is divided evenly among the base station's antennas.
- (ii) The transmit power budget is divided among the base station's antennas, according to the channel state information feedback provided by the mobile phone to the base station.
- (iii) No transmit power is assigned to the user if it experiences a bad channel condition.



(i) The transmit power budget is divided evenly among the base station's antennas.

→no up-to-date channel state information can be fed back to the transmitter if users are moving very fast, as channel coherence time is shorter than feedback+transmission time.

题目三(飞机场场景,时速1000km/h,基站多天线下行功率分配): 正确答案是 (i) 平均分配功率给各天线。原因是用户移动速度太快时,信道相干时间(channel coherence time,即信道特性保持基本不变的时间窗口)比"反馈信道状态信息 + 传输"所需的时间还短——等基站收到你反馈的信道信息,信道早就变了,所以这时候用信道状态信息来做智能功率分配(选项ii)反而没用,不如干脆平均分配。

Counterexample: Spectrum re-farming of TV bands due to efficiencies in the switch to digital TV

这句话是在给"频谱很难从持牌用户手里收回"这个说法举一个反例——也就是说,虽然通常很难收回频谱,但确实发生过一次成功的例子。

背景:

模拟电视信号传输效率很低,占用了很宽的频段(比如之前提到的 VHF/UHF 那些频段:54-88 MHz, 174-216 MHz, 470-806 MHz)。

发生了什么(Spectrum re-farming, 频谱重耕/重新分配):

各国推动电视从模拟信号切换到数字信号(Digital TV)。数字电视信号压缩效率高得多,同样的画质只需要占用小得多的频段。

于是切换到数字电视后,之前模拟电视占用的一大块频谱就被"腾出来"了——这些原本分配给电视广播的频段,被政府收回,重新分配给其他用途(比如 4G/LTE 移动通信)。这个"收回旧用途的频段、重新分配给新用途"的过程就叫 spectrum re-farming(频谱重耕)。

为什么这是个"反例"(counterexample):

前一句话说"很难从持牌者手里把频谱收回",但电视这个案例恰恰是成功收回并重新分配的少数例子——之所以能成功,是因为:

技术进步(模拟转数字)让原有用户可以用更少的频谱做同样的事,腾出"富余"的部分。

这是政府主导、有明确技术路线图的转型(不是强行从持牌人手里抢频谱,而是"以旧换新",电视台照样能播,只是用的频段少了)。

所以这句话的意思就是:"频谱难收回"不是绝对的,数字电视替代模拟电视就是一个成功收回并重新利用频谱的典型例子。